Quadratics: Expanding & Factorising

Practice test 6 — answer sheet · 40 marks

Section A — Expanding double brackets

A1. $x^2 + 12x + 5x + 60$ (1) = $\mathbf{x^2 + 17x + 60}$ (1)

A2. $x^2 - 11x - 3x + 33$ (1) = $\mathbf{x^2}$ – **14x + 33** (1)

A3. $(3x - 4)(3x - 4) = 9x^2 - 12x - 12x + 16$ (1) = $\mathbf{9x^2}$ – **24x + 16** (1)

A4. 7 – ? = 2, so the first missing number is **5** (1) 7 × 5 = 35, so the second is **35** (1)

Section B — Expanding three brackets

B1. $(x - 3)(x - 2) = x^2 - 5x + 6$ (1) $(x^2 - 5x + 6)(x + 6) = x^3 + 6x^2 - 5x^2 - 30x + 6x + 36$ (1) = $\mathbf{x^3 + x^2}$ – **24x + 36** (1)

B2. $(x - 3)^2 = x^2 - 6x + 9$ (1) $(x^2 - 6x + 9)(x - 3) = x^3 - 3x^2 - 6x^2 + 18x + 9x - 27$ (1) = $\mathbf{x^3}$ – $\mathbf{9x^2}$ + **27x** – **27** (1)

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Multiply to –54, add to 3: 9 and –6 (1) **(x + 9)(x** – **6)** (1)

C2. Multiply to 72, add to –17: –8 and –9 (1) **(x** – **8)(x** – **9)** (1)

C3. $(4x)^2 - 5^2$ = **(4x + 5)(4x** – **5)** (1)

C4. $7(x^2 - 1)$ (1) = **7(x + 1)(x** – **1)** (1)

Section D — Factorising $ax^2 + bx + c$

D1. 2 × 10 = 20; numbers: –4 and –5, so $2x^2 - 4x - 5x + 10 = 2x(x - 2) - 5(x - 2)$ (1) **(2x** – **5)(x** – **2)** (1)

D2. 4 × –3 = –12; numbers: 6 and –2, so $4x^2 + 6x - 2x - 3 = 2x(2x + 3) - 1(2x + 3)$ (1) **(2x** – **1)(2x + 3)** (1)

D3. 9 × 4 = 36; numbers: –6 and –6, so $9x^2 - 6x - 6x + 4 = 3x(3x - 2) - 2(3x - 2)$ (1) $\mathbf{(3x - 2)^2}$ (1)

Section E — Using factorising

E1. $x^2 + 7x - 18 = 0$ (1) $(x + 9)(x - 2) = 0$ (1) **x = –9 or x = 2** (1)

E2. Numerator = (2x – 1)(x + 3) (1) denominator = (2x + 1)(2x – 1) (1) cancel (2x – 1): **(x + 3) / (2x + 1)** (1)

Section F — Problem solving & proof

F1. (a) $(2x + 3)(x - 1) = 2x^2 + x - 3$ (1) $2x^2 + x - 3 = 18$, so $2x^2 + x - 21 = 0$ (1)
(b) (2x + 7)(x – 3) = 0 so x = 3 or x = –3.5; a length can't be negative, so x = 3 (1) the rectangle is 9 cm by 2 cm, so **perimeter = 22 cm** (1)

F2. $(4x^2 + 12x + 9) - (4x^2 - 12x + 9)$ (1) = $4x^2 + 12x + 9 - 4x^2 + 12x - 9$ = **24x** (1)

F3. (9.6 + 0.4)(9.6 – 0.4) = 10 × 9.2 = **92** (1)

Marking note: the marks in brackets show where each mark is earned. Award method marks for a correct method even if there is a later slip. Accept answers in any equivalent form, e.g. terms in a different order, or brackets written the other way round: (x – 7)(x + 4) is the same as (x + 4)(x – 7). "Factorise fully" questions lose the last mark if a common factor is left inside a bracket.